

Beginning Algebra Final Test (Part I) — Practice Test 2

Instructions

- i) As always, write a title, name, date, and course.
- ii) Number each problem, circle the number, and show work, even when you can do the problems in your head.
- iii) Draw a rectangle around your answers.
- iv) Staple a copy of these test questions with your work and your final answers.

Solve.

$$\textcircled{1} 0 = 4b^3 - 8b^2 - 9b + 18$$

$$\textcircled{2} 4x^2 = 16x$$

$$\textcircled{3} 25b^4 + 4 = 101b^2$$

$$\textcircled{4} y^{10} - 4y^9 = 12y^8$$

Perform the indicated operations and simplify completely.

$$\textcircled{5} \frac{2ab - 8b^2}{9ab} \cdot \frac{3a^2 - 75b^2}{a^2 + ab - 20b^2}$$

$$\textcircled{6} \frac{5}{3x+15} + \frac{2}{5x+25}$$

$$\textcircled{7} \frac{3}{x-2} - \frac{x+6}{3x-1}$$

$$\textcircled{8} \frac{40x^3}{5x^6-15x^4-20x^2} \div \frac{x}{2-x}$$

- $\textcircled{9}$ a) Simplify the following polynomial.
 b) Write the polynomial in descending order of powers.
 c) State the polynomial's degree.

$$4x + 2x^3 - 1 - 2 - 15x^3 + x^2 - 5x^2 + 7x - 7$$

d) Expand: $(8x-1)^2$

Perform the indicated operations and simplify completely.

⑩ Divide: $\frac{-14x^8 - 28x^5 + 49x^3}{-7x^3}$

⑪ Expand: $(x-6)(-2x^2+7x-1)$

⑫ a) Add: $(7x^4 + 2x^3 - 4x^2 - 6x + 1) + (8x^3 - 2x + 10)$

b) Subtract: $(x^2 - 2x + 4) - (x^2 + x - 8)$

⑬ Combine.

$$\frac{4}{x} - \frac{2}{x+3} + \frac{3x+1}{x^2+3x}$$

⑭ Simplify:

$$\frac{24x^3y - 8x^2y^2}{3x^2 - 7xy + 2y^2}$$

Rationalize the denominators.

⑮ $\sqrt{\frac{7}{64}}$

⑯ $\sqrt{\frac{11}{20}}$

⑰ $\frac{1}{\sqrt{2}}$

⑱ $\sqrt{\frac{32}{45}}$

⑲ Simplify the expressions.

a) $\sqrt{100}$

b) $2\sqrt{22}(-3\sqrt{2})$

$$c) \sqrt{\frac{9}{16}}$$

$$d) 9\sqrt{45} - 2\sqrt{25} - \sqrt{80}$$

Simplify.

$$\textcircled{20} \frac{2y^3}{\frac{x}{4y}}$$

$$\textcircled{21} \frac{\frac{1}{3x^2+6x} + \frac{-3}{x+2}}{\frac{8}{3} - \frac{1}{x}}$$

For the problems below, always label the axes of your graphs and the intervals on each axis. Write all coordinates you use to make your graphs (e.g. (2,-3)).

②② Graph the following equation using the slope and y-intercept.

$$-\frac{3}{5}x - 2 = y$$

②③ Graph the equation using slope and a point visible in the equation.

$$y - 1 = 4(x - 1)$$

②④ Graph the following equation using x and y intercepts.

$$-8 = 4x - 5y$$

②⑤ Graph the following equation using any method.

$$3 - 2y = 2x$$

Beginning Algebra Final Test (Part I) - Practice Test 2 Answers

① $b = \frac{3}{2}, -\frac{3}{2}, \text{ or } 2$

② $x = 0 \text{ or } 4$

③ $b = \frac{1}{5}, -\frac{1}{5}, 2, \text{ or } -2$

④ $y = 0, 6, \text{ or } -2$

⑤ $\frac{2(a-5b)}{3a} \text{ or } \frac{2a-10b}{3a}$

⑥ $\frac{31}{15(x+5)} \text{ or } \frac{31}{15x+75}$

⑦ $\frac{-x^2+5x+9}{(x-2)(3x-1)} \text{ or } \frac{-x^2+5x+9}{3x^2-7x+2}$

⑧ $\frac{-8}{(x+2)(x^2+1)} \text{ or } \frac{-8}{x^3+2x^2+x+2}$

⑨ a & b) $-13x^3 - 4x^2 + 11x - 10$

c) $D = 3$

d) $64x^2 - 16x + 1$

⑩ $2x^5 + 4x^2 - 7$

⑪ $-2x^3 + 19x^2 - 43x + 6$

⑫ a) $7x^4 + 10x^3 - 4x^2 - 8x + 11$

b) $-3x + 12$

⑬ $\frac{5x+13}{x(x+3)} \text{ or } \frac{5x+13}{x^2+3x}$

⑭ $\frac{8x^2y}{x-2y}$

⑮ $\frac{\sqrt{7}}{8}$

⑯ $\frac{\sqrt{55}}{10}$

⑰ $\frac{\sqrt{2}}{2}$

⑱ $\frac{4\sqrt{10}}{15}$

⑲ a) 10 b) $-12\sqrt{11}$ c) $\frac{3}{4}$ d) $23\sqrt{5} - 10$

20 $\frac{8y^4}{x}$

21 $\frac{1-9x}{8x^2+13x-6}$ or $\frac{1-9x}{(x+2)(8x-3)}$

